Scalability & Future Plan

Date	11 July 2024
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	On-Device Model Latency	Running the LaMini-Flan-T5 pipeline locally uses significant CPU/RAM, blocking the FastAPI thread and slowing down explanations.	High
2	Ephemeral Session State	Streamlit session state (st.session_state) is held in-memory. If the browser is refreshed, all user context and quiz results are lost.	High
3	Unprotected API Routing	The FastAPI backend endpoints do not have user authentication or rate-limiting middleware, exposing backend APIs to abuse.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs locally as a single-instance Streamlit app communicating over localhost:8000	Containerize the Streamlit and FastAPI apps into Docker images, deploying them on AWS ECS behind an Application Load Balancer (ALB).
2	Data Storage	No database layer; data is loaded transiently or kept in Streamlit session variables.	Integrate a PostgreSQL relational database managed on AWS RDS to persistently store user profiles, generated quizzes, and saved history.
3	Performance	Local model inference runs directly in the server process, causing significant request queue bottlenecks.	Offload the LaMini pipeline to serverless GPU containers (like RunPod) or migrate to the cloud-hosted Hugging Face Inference API.
4	Security	Gemini API keys are read from a local .env configuration file.	Store API credentials in AWS Secrets Manager and restrict API endpoints using JWT (JSON Web Tokens) user authentication.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	MVP Deployment & Core Refinement	1 Month	Establishes the core hybrid architecture by deploying the first 5 fully operational modules (Q&A, Quiz, Explainer, Summarizer, and Dashboard).
Phase 2	Database Integration & Auth	2 Months	Enables user accounts, allows students to save their history, and resolves the rate-limiting on the Personalized Learning Path module.
Phase 3	Serverless AI Execution	2 Months	Decreases response time for the Concept Explainer from ~15 seconds to under 1.5 seconds.
Phase 4	Collaborative Study Rooms	4 Months	Allows students to share real-time interactive quizzes, summaries, and learning roadmaps peer-to-peer.